

Fourier Localization, Sampling, and Frequency Resolution

1 One frequency and perfect time localization

We use the continuous Fourier transform convention

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-i2\pi f t} dt, \quad x(t) = \int_{-\infty}^{\infty} X(f) e^{i2\pi f t} df,$$

where f is measured in hertz.

A single-frequency complex exponential has the transform pair

$$\boxed{e^{i2\pi f_0 t} \longleftrightarrow \delta(f - f_0).}$$

Its frequency is perfectly specified, but it extends indefinitely in time. A real sinusoid contains a positive and a negative frequency:

$$\cos(2\pi f_0 t) \longleftrightarrow \frac{1}{2} [\delta(f - f_0) + \delta(f + f_0)].$$

It is likewise completely delocalized in time.

The opposite ideal limit is a Dirac impulse:

$$\boxed{\delta(t - t_0) \longleftrightarrow e^{-i2\pi f t_0}.}$$

The impulse is concentrated at one instant. Its spectrum contains every frequency with equal magnitude and phases arranged to concentrate the signal at t_0 . For $t_0 = 0$, the spectrum is simply $X(f) = 1$.

All frequencies alone are insufficient. Their relative amplitudes and phases determine the resulting signal. A Gaussian, for example, has nonzero spectral amplitude at every finite frequency but is not an impulse.

These are ideal limits, interpreted using distributions. Neither an unlimited sinusoid nor a Dirac impulse is a finite-energy signal, so we do not directly apply finite-energy standard-deviation formulas to them.

A single frequency is completely delocalized in time. Perfect localization at one instant requires an unlimited frequency range with appropriate amplitudes and phases.

2 The Fourier uncertainty principle

For a nonzero finite-energy signal, Parseval's theorem gives

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\infty}^{\infty} |X(f)|^2 df.$$

Define the energy centroids and variances by

$$\begin{aligned} \bar{t} &= \frac{1}{E} \int_{-\infty}^{\infty} t |x(t)|^2 dt, & \bar{f} &= \frac{1}{E} \int_{-\infty}^{\infty} f |X(f)|^2 df, \\ \sigma_t^2 &= \frac{1}{E} \int_{-\infty}^{\infty} (t - \bar{t})^2 |x(t)|^2 dt, & \sigma_f^2 &= \frac{1}{E} \int_{-\infty}^{\infty} (f - \bar{f})^2 |X(f)|^2 df. \end{aligned}$$

When both variances are finite, the Fourier uncertainty relation is

$$\sigma_t \sigma_f \geq \frac{1}{4\pi}.$$

For angular frequency $\omega = 2\pi f$, it becomes $\sigma_t \sigma_\omega \geq 1/2$.

This is a mathematical property of Fourier pairs, independent of sampling. A signal cannot have arbitrarily small energy spreads in both domains. The infinite extent of Fourier basis functions provides intuition, while the precise bound follows from the Fourier relationship between the two representations. A Gaussian reaches equality.

3 Sampling and finite records

Use δt and δf for sampling intervals, reserving σ_t and σ_f for signal energy spreads.

For N samples at $t_n = n\delta t$, $n = 0, \dots, N-1$, the standard DFT uses

$$\boxed{T = N\delta t}, \quad \boxed{\delta f = \frac{1}{T}}, \quad \boxed{f_N = \frac{1}{2\delta t}}.$$

Here T is the DFT period or record duration; the elapsed time between the first and last sample is $(N-1)\delta t$.

Change	Consequence
Smaller δt	Larger frequency range available without aliasing.
Longer DFT record T	Smaller frequency-bin spacing δf .
Longer actual observation	Can improve discrimination of nearby frequencies.

Avoiding aliasing requires spectral content below the Nyquist limit, normally enforced with suitable filtering before sampling. The full DFT frequency interval has width $1/\delta t = 2f_N$.

Sampling imposes representation constraints rather than a second intrinsic uncertainty principle. Frequency-bin spacing is also not identical to resolving power: distinguishing nearby components depends on observation duration, window shape, noise, and the estimation method.

4 Observation windows connect the two ideas

Observing through a finite window $w(t)$ gives

$$x_{\text{obs}}(t) = x(t)w(t) \quad \longleftrightarrow \quad X_{\text{obs}}(f) = (X * W)(f),$$

where $*$ denotes convolution in frequency. For a rectangular window centered at zero with duration T_{obs} ,

$$W(f) = T_{\text{obs}} \text{sinc}(fT_{\text{obs}}), \quad \text{sinc}(u) = \frac{\sin(\pi u)}{\pi u}, \quad \text{sinc}(0) = 1.$$

Its first zeros occur at $f = \pm 1/T_{\text{obs}}$, so the main-lobe width between those zeros is $2/T_{\text{obs}}$. A shorter observation window therefore broadens spectral features; a longer one narrows them.

Why zero-padding does not improve resolving power

Appending zeros increases the duration used by the DFT and decreases δf . It adds no observations and does not narrow the observation window's spectral main lobe. It samples the existing spectrum more densely.

For example, 1000 samples with $\delta t = 0.001$ s give $T = 1$ s, $\delta f = 1$ Hz, and $f_N = 500$ Hz. Padding to 4000 samples changes the DFT spacing to 0.25 Hz, but the actual observation remains 1 s. Recording 4 s of actual data at the same interval can improve frequency discrimination; appending 3 s of zeros does not provide that information.

Fourier uncertainty concerns how concentrated a signal can be in time and frequency. Sampling determines the frequency range we can represent, while observation duration helps determine how well we can distinguish nearby frequencies. These are distinct ideas, connected by finite observation windows.